FIVE EQUIVALENT CHARACTERIZATIONS OF UNIVERSALLY INJECTIVE UNRAMIFIED MORPHISMS

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Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original reference 02HE: sources/stacks-chow/tags.txt, source lines 3261–3261. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02GE: sources/stacks-chow/tags.txt, source lines 3225–3225. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01S4: sources/stacks-chow/tags.txt, source lines 2325–2325. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02YC: sources/stacks-chow/tags.txt, source lines 3871–3871. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03DP: sources/stacks-chow/tags.txt, source lines 4428–4428. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02G8: sources/stacks-chow/tags.txt, source lines 3219–3219. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. The following are equivalent:*

1. *f is unramified and a monomorphism,*
2. *f is unramified and universally injective,*
3. *f is locally of finite type and a monomorphism,*
4. *f is universally injective, locally of finite type, and formally unramified,*
5. *f is locally of finite type and X_s is either empty or $X_s \to s$ is an isomorphism for all $s \in S$.*

Proof. We have seen in More on Morphisms, Lemma 02HE that being formally unramified and locally of finite type is the same thing as being unramified. Hence (4) is equivalent to (2). A monomorphism is certainly universally injective and formally unramified hence (3) implies (4). It is clear that (1) implies (3). Finally, if (2) holds, then $\Delta : X \to X \times_S X$ is both an open immersion (Morphisms, Lemma

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02GE) and surjective (Morphisms, Lemma 01S4) hence an isomorphism, i.e., f is a monomorphism. In this way we see that (2) implies (1).

Condition (3) implies (5) because monomorphisms are preserved under base change (Schemes, Lemma 02YC) and because of the description of monomorphisms towards the spectra of fields in Schemes, Lemma 03DP. Condition (5) implies (4) by Morphisms, Lemmas 01S4 and 02G8. □